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A Annotations in popular datasets

Many popular machine learning datasets have been created using annotators to determine labels. Often, they also use very basic aggregation and voting mechanisms (corresponding to $\mathbb{P}_{\text{agg}}$ and $\mathbb{P}_{\text{vote}}$) such as majority voting or averaging. For example, the list of most cited datasets on Papers with Code[7] includes the following:

- The CIFAR10 dataset (Krizhevsky, 2009) is a popular multiclass classification benchmark of 32×32 pixel color images. It was labeled using a strategy similar to majority voting: first, a crowd-sourced label was collected which was then verified and potentially corrected by an author.
- CIFAR10H (Peterson et al., 2019) revisited the CIFAR10 dataset by gathering additional human annotations where each annotator provides a single label.
- ImageNet (Russakovsky et al., 2015) uses multiple annotators with majority voted labels.
- COCO (Lin et al., 2014) labels a category as present if any out of 8 annotators labeled it as present (even if all other annotators disagree).
- The Stanford sentiment treebank (Socher et al., 2013) includes 3 annotations per example which are used in various ways during evaluation.
- MultiNLI (Williams et al., 2018) includes 5 annotations which are majority voted.
- The semantic textual similarity benchmark (Cer et al., 2017) averages scores across multiple annotators.
- The Recognizing Textual Entailment datasets[8] also consider multiple annotators and several editions of the corresponding challenge simply neglected examples with disagreeing annotators (equivalent to majority voting while ignoring ambiguous examples).

The above natural language processing datasets are all part of the GLUE benchmark (Wang et al., 2019) – the go-to benchmark for natural language understanding. Datasets beyond (multi-label) classification also often include multiple annotations. VQA (Goyal et al., 2017), for example, includes 10 free-form answers per question; aggregating them is clearly non-trivial. For many datasets, it might also be unclear how annotations have been collected or aggregated. The Quora Question Pairs webpage[9] explicitly mentions label errors but does not give details on annotation and aggregation. All of these examples emphasize that aggregating annotations and using majority voted or averaged labels is extremely common across supervised machine learning. This means that plausibilities are typically readily obtainable and the core problem we address – using majority voted labels on ambiguous tasks for calibration leads to under-coverage – is highly relevant.

B Calibration threshold and p-values

B.1 Single p-value

We include for completeness a proof of the results presented in Section 2.1. These are standard results. The p-value formulation detailed here will be then extended to establish the validity of some of the procedures proposed in this work.

We first establish that the prediction set defined by Equations (2) and (3) satisfies

$$1 - \alpha \leq \mathbb{P}(Y \in C(X)) \leq 1 - \alpha + \frac{1}{n+1}, \tag{20}$$

the upper bound requiring the additional assumption that the conformity scores are almost surely distinct.

[7] https://paperswithcode.com/datasets?q=&v=lst&o=cited
[8] https://tac.nist.gov//2008/rte/past_data/index.html
[9] https://quoradata.quora.com/First-Quora-Dataset-Release-Question-Pairs

Let us write $(X_{n+1}, Y_{n+1}) = (X, Y)$ and $S_i = E(X_i, Y_i)$. As $(S_i)_{i\in[n+1]}$ are exchangeable, then a direct application of (Romano et al., 2019, Appendix A, Lemma 2) shows that for any $\alpha \in (0,1)$ and $\tau' = Q\left(\{-S_i\}_{i\in[n]}; \lceil(1-\alpha)(n+1)\rceil/n\right)$

$$\mathbb{P}(-S_{n+1} \leq \tau') \geq 1 - \alpha \tag{21}$$

and, if the random variables $(S_i)_{i\in[n+1]}$ are almost surely distinct, then

$$\mathbb{P}(-S_{n+1} \leq \tau') \leq 1 - \alpha + \frac{1}{n+1}. \tag{22}$$

So $C(X_{n+1})$ includes all the values y such that $-S(X_{n+1}, y)$ is smaller or equal than the $\lceil(1-\alpha)(n+1)\rceil$ smallest values of $(-S_i)_{i\in[n]}$. This is equivalent to considering all the values y such that $S(X_{n+1}, y)$ is larger or equal than the $\lfloor\alpha(n+1)\rfloor$ smallest values of $(S_i)_{i\in[n]}$. Hence this corresponds to the prediction set defined by Equations (2) and (3).

We now show that this prediction set can also be obtained by thresholding p-values. Indeed Equation (4) can be rewritten as

$$\rho_{Y_{n+1}} = \frac{\sum_{i=1}^{n+1} \mathbb{I}(S_i \leq S_{n+1})}{n+1}. \tag{23}$$

As $(S_i)_{i\in[n+1]}$ are exchangeable, then $\rho_{Y_{n+1}}$ is uniformly distributed on $\{\frac{1}{n+1}, \frac{2}{n+1}, ..., 1\}$ if the distribution of the scores is continuous and thus $\rho_{Y_{n+1}}$ is a p-value, i.e. $\mathbb{P}(\rho_{Y_{n+1}} \leq \alpha) \leq \alpha \iff \mathbb{P}(\rho_{Y_{n+1}} > \alpha) \geq 1 - \alpha$. It can be checked that this property still holds if the distribution of the scores is not continuous (see e.g. (Bates et al., 2023)). So if we define the prediction set as

$$C(X_{n+1}) = \{y : \rho_y > \alpha\} \tag{24}$$

then by construction it follows that

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha. \tag{25}$$

We show here that Equation (24) does indeed coincide with the set defined by Equations (2) and (3). We have

$$\rho_y > \alpha \iff \sum_{i=1}^{n} \mathbb{I}(S_i \leq E(X_{n+1}, y)) > \alpha(n+1) - 1. \tag{26}$$

If $\alpha(n+1)$ is an integer then $\sum_{i=1}^{n} \mathbb{I}(S_i \leq E(X_{n+1}, y)) \geq \alpha(n+1)$, i.e. $E(X_{n+1}, y)$ is larger or equal than the $\alpha(n+1)$ smallest values of $(S_i)_{i\in[n]}$. If $\alpha(n+1)$ is not an integer then it means that $\sum_{i=1}^{n} \mathbb{I}(S_i \leq E(X_{n+1}, y)) \geq \lceil\alpha(n+1)\rceil - 1$, i.e. $E(X_{n+1}, y)$ is larger than the $\lceil\alpha(n+1)\rceil - 1$ smallest values of $(S_i)_{i\in[n]}$. However, we have $\lceil\alpha(n+1)\rceil - 1 = \lfloor\alpha(n+1)\rfloor$ as $\alpha(n+1)$ is not an integer. So overall, we have that $\rho_y > \alpha$ corresponds to $E(X_{n+1}, y)$ being larger or equal than the $\lfloor\alpha(n+1)\rfloor$ smallest values $(S_i)_{i\in[n]}$. So the prediction set in Equation (24) does coincide with the set defined by Equations (2) and (3).

Finally note that when the distributions of the scores is continuous, i.e. the scores are almost surely distinct, then we also have $\mathbb{P}(\rho_{Y_{n+1}} \leq \alpha) \geq \alpha - 1/n+1$ so $\mathbb{P}(\rho_{Y_{n+1}} > \alpha) \leq 1 - \alpha + 1/n+1$.

B.2 Average of p-values

We establish here the expression of the prediction set obtained by thresholding the following average p-value

$$\rho_{Y_{n+1}} = \frac{\sum_{j=1}^{m} \sum_{i=1}^{n+1} \mathbb{I}(S_i^j \leq S_{n+1}^j)}{m(n+1)} \tag{27}$$

where $S_{n+1}^j = S_{n+1}$ for all $j \in [m]$. We know that by construction the set $C(X_{n+1}) = \{y : \rho_y > \alpha\}$ is such that

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - 2\alpha. \tag{28}$$